

MySQL

Files(Notepad file, ms word, excel,) ==> storing the data / information
i want to store the details of all 100 students
Name, age, city, branch

Files are good for reading, updating or deleting ==> less data
i want to store huge amount of data(all students of CSE branch) ==> Databases

File ==> water bottle
Database ==> water tank

I want to store details of 100 students

- 1) create a database
- who will create database ==> softwares(MySQL, Oracle, MongoDB,)
- MySQL, Oracle,..... ==> softwares ==> create the databases, delete the database, update the database
- database ==> employee_db ==> emp_db
- Computer ==> Install MySQL
- 1) create a database **books db**
2) store the details of 5 books(insert operation)

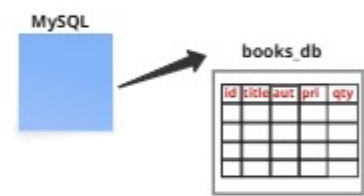


Table format

<https://github.com/uday5252/Java-Full-Stack-Batch-2.git>

Practice Data: `student_library`

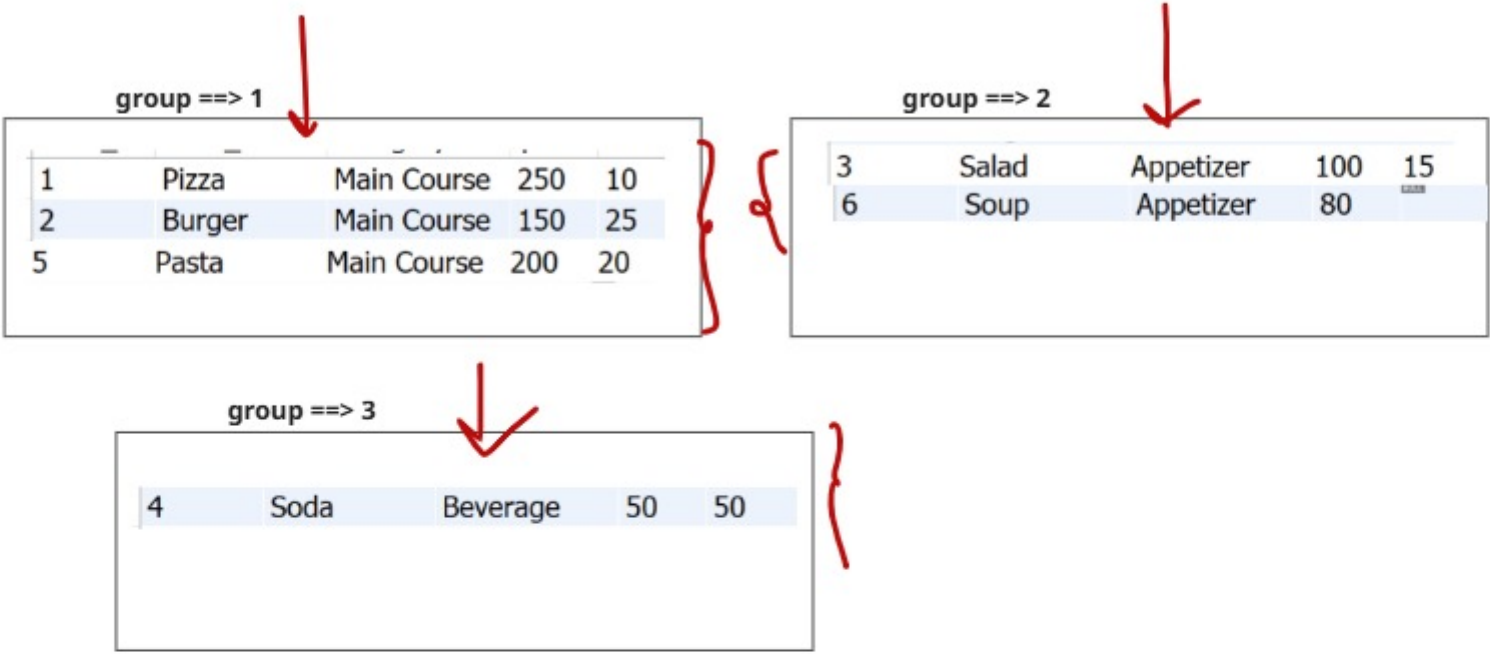
student_id	name	department	fine_amount
1	Rahul	Computer	100
2	Priya	Science	200
3	Amit	Computer	50
4	Sneha	Arts	300
5	Vijay	NULL	150
6	Kiran	Science	400

4) Display all students
who belong to the 'Science' department
and have a fine_amount greater than 250.

- 1) Display all students whose fine_amount is not between 100 and 300.
2) Display all students who do not belong to 'Science' or 'Arts' departments.
3) Display all students whose name ends with the letter 'a'.
`select * from students_library where name like "%a"`
- 5) Display all students whose name has exactly 4 letters (Hint: use the underscore _ character).

aggregate functions, group by, having

food_id	food_name	category	price	stock
1	Pizza	Main Course	250	10
2	Burger	Main Course	150	25
3	Salad	Appetizer	100	15
4	Soda	Beverage	50	50
5	Pasta	Main Course	200	20
6	Soup	Appetizer	80	



`select category, sum(price) from menu group by category;`

Main Course
Appetizer
Beverage

having ==> will be used after group by

where condtion
group by
having condition

Airline Data Table: `FLights`

Flight_ID	Departure_City	Destination	Price
101	New York	London	500
102	New York	Paris	450
103	London	Paris	200
104	London	Dubai	600
105	New York	Dubai	700
106	Paris	Dubai	550
107	London	New York	500

- 1) Total number of flights departing from each city
- 2) Cities with 3 or more flights